

Uncertainty-Aware AI-Assisted Video Review for Faster and More Reliable Officiating in Competitive Taekwondo

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Introduction

Instant Video Replay (IVR) is indispensable for resolving contested scoring actions in Taekwondo, yet manual review can interrupt competition flow and requires officials to interpret brief, high-speed interactions under substantial time pressure. Head kicks are particularly demanding because the scoring decision depends on contact validity, action type, and, for turning techniques, correct technical classification. We investigate whether FST.ai and its uncertainty-aware extension, FST.ai 2.0, can support faster and more transparent review while preserving the referee as the final decision-maker.

Methods

FST.ai is a real-time decision-support framework for head-kick review that combines computer vision, pose estimation, motion classification, impact-related evidence, and edge inference. FST.ai 2.0 extends this architecture with pose-based action recognition using graph convolutional networks, epistemic-uncertainty representation through credal sets, and visual explanation overlays. The evaluation uses annotated competition video and expert officiating information to assess review latency, action recognition and classification, agreement with expert decisions, and the interaction between AI-generated evidence and human judgment. The design is explicitly human-in-the-loop: model outputs are recommendations accompanied by interpretable evidence and uncertainty rather than autonomous scoring decisions.

Results

The original FST.ai study demonstrated that the head-kick review workflow can move from conventional IVR delays measured in tens of seconds toward seconds-scale AI-assisted decision support. Subsequent competition-data validation reported for FST.ai 2.0 showed an 85% reduction in decision-review time and 93% referee trust in AI-assisted decisions. Beyond speed, the system exposes pose, action, and uncertainty evidence to the official, allowing low-confidence or conflicting cases to remain under human adjudication instead of forcing a binary machine decision.

Conclusion

These results support a model of sports analytics in which AI augments rather than replaces officiating. For Taekwondo, the relevant performance objective is not prediction accuracy alone, but the joint quality of decision reliability, review latency, interpretability, and appropriate human deferral under epistemic uncertainty. FST.ai therefore provides a practical framework for uncertainty-aware, auditable video review in high-speed competition and a basis for studying how human–AI decision support can improve officiating without transferring final authority away from the referee.

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